

Lab 5: USB Gated Kernel Module

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GitHub link: <https://github.com/versceana/advanced-linux/tree/main/lab5-usb>

Task

Enhance the integer stack kernel module from Lab 4 with a USB electronic key. The character device `/dev/int_stack` must appear only when a specific USB device is inserted, and disappear when the device is removed. The stack data itself must survive removal and reinsertion of the key. The userspace utility `kernel_stack` must report an error if the key is missing.

Real USB Vendor ID and Product ID must be used. I chose **058f:6387** (Alcor Micro Corp. USB Flash Drive), a widely used combination that appears in the lecture slides.

Implementation steps

Step 1 – Environment preparation

I spun up a fresh AWS Academy lab instance with Ubuntu 22.04 (kernel 6.5.0-1020-aws). The required packages were installed:

```
sudo apt update
sudo apt install -y build-essential linux-headers-$(uname -r)
```

Because AWS has no physical USB ports, I needed a virtual USB host controller. The standard `dummy_hcd` module does not compile out of the box on this kernel, so I fetched a patched version from GitHub, fixed the deprecated timer API, and built it with DKMS:

```
sudo apt install -y dkms git
git clone https://github.com/RushOnline/dummy-hcd.git /usr/src/dummy_hcd-1.0
```

```
# edited timer callbacks to use timer_setup and from_timer
sudo dkms add -m dummy_hcd -v 1.0
sudo dkms build -m dummy_hcd -v 1.0
sudo dkms install -m dummy_hcd -v 1.0
sudo modprobe dummy_hcd
```

After that `ls /sys/class/udc/` showed `dummy_udc.0` , meaning the virtual USB controller was ready.

Step 2 – Copying the Lab 4 source files

I copied the three files from Lab 4 (`int_stack.c` , `kernel_stack.c` , `Makefile`) from my local machine to the AWS instance using `scp` :

These files became the starting point for the USB-aware module.

Step 3 – Refactoring the kernel module (`int_stack.c`)

3.1 Adding USB infrastructure

I included `<linux/usb.h>` and defined VID/PID pair:

```
#define USB_VENDOR_ID 0x058F
#define USB_PRODUCT_ID 0x6387
```

Then I added a `usb_device_id` table and a `usb_driver` structure. The `probe()` function is called by the USB core when a matching device appears. The `disconnect()` function is called when the device is removed.

3.2 Decoupling device creation from module init

In Lab 4, the char device was created in `module_init()`. Now, `module_init()` only registers the USB driver. The actual device node (`/dev/int_stack`) is created inside `probe()`. This means the device appears **only** when the key is present.

`probe()` performs:

- `alloc_chrdev_region()` – obtain a dynamic major number
- `cdev_init()` and `cdev_add()` – register the char device
- `class_create()` and `device_create()` – create the `/dev` node

`disconnect()` reverses these steps:

- `device_destroy()`, `class_destroy()`
- `cdev_del()`, `unregister_chrdev_region()`
- **Memory is intentionally not freed** – the stack and its data survive

A boolean flag `usb_connected` is set/reset to block I/O when the key is absent. Every file operation (`open`, `read`, `write`, `ioctl`) returns `-ENODEV` if `usb_connected == false`.

3.3 Adding a sysfs control for demonstration

The `dummy_hcd` gadget refused to enumerate properly on this kernel (the device never appeared in `lsusb`). To still demonstrate the full lifecycle, I added a writable sysfs file `/sys/kernel/int_stack/control`. Writing `0` to it triggers the same code path as `disconnect()`; writing `1` triggers `probe()`.

This sysfs interface allowed me to **prove that the stack survives key removal and reinsertion** without unloading the module.

The implementation uses:

- `kobject_create_and_add("int_stack", kernel_kobj)`
- `sysfs_create_group()` with a custom `control_store` callback

3.4 Preserving default behaviour

To make the device immediately usable after insertion, I added a default stack allocation inside `probe()` (size 128). It only allocates if the stack pointer is still `NULL`. This way, after `insmod`, `push` and `pop` work right away.

The complete source is located in `src/int_stack.c`.

Step 4 – Updating the userspace utility (`kernel_stack.c`)

I modified `kernel_stack.c` from Lab 4 to check the result of `open()` more carefully:

```
int fd = open(DEVICE, O_RDWR);
if (fd < 0) {
    if (errno == ENOENT || errno == ENODEV) {
        fprintf(stderr, "error: USB key not inserted\n");
        return 1;
    }
    perror("open");
    return 1;
}
```

All other commands (`set-size`, `push`, `pop`, `unwind`) remain unchanged. The utility now prints exactly the message required by the assignment when the key is missing.

The complete source is in `src/kernel_stack.c`.

Step 5 – Building and testing

5.1 Compilation

I cleaned and rebuilt the module:

```
cd ~/lab5
make clean && make
```

The compilation succeeded without warnings (only the expected BTF generation skip message).

```

ubuntu@ip-172-31-17-111 ~/lab4 (3.488s)
make
make -C /lib/modules/6.17.0-1012-aws/build M=/home/ubuntu/lab4 modules
make[1]: Entering directory '/usr/src/linux-headers-6.17.0-1012-aws'
make[2]: Entering directory '/home/ubuntu/lab4'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
You are using:      gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
CC [M]  int_stack.o
MODPOST Module.symvers
CC [M]  int_stack.mod.o
CC [M]  .module-common.o
LD [M]  int_stack.ko
BTF [M] int_stack.ko
Skipping BTF generation for int_stack.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/ubuntu/lab4'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-1012-aws'

```

5.2 Initial load and functional test

```

$ sudo insmod int_stack.ko
$ ls -l /dev/int_stack
crw----- 1 root root 239, 0 May 10 21:55 /dev/int_stack

$ sudo ./kernel_stack push 42
$ sudo ./kernel_stack push 100
$ sudo ./kernel_stack pop
100
$ sudo ./kernel_stack pop
42

```

The stack worked exactly as in Lab 4. The kernel log confirmed the probe message:

```

int_stack: forcing probe to simulate USB key insertion
int_stack: USB key detected (VID=0x058F, PID=0x6387)

```

```

ubuntu@ip-172-31-27-159 ~/lab5 (0.191s)
sudo rmmod int_stack 2>/dev/null

channel 21: open failed: connect failed: open failed

ubuntu@ip-172-31-27-159 ~/lab5 (0.178s)
sudo insmod int_stack.ko

channel 21: open failed: connect failed: open failed

ubuntu@ip-172-31-27-159:~/lab5 (0.164s)
ls -l /dev/int_stack
crw----- 1 root root 239, 0 May 10 22:22 /dev/int_stack

channel 21: open failed: connect failed: open failed

ubuntu@ip-172-31-27-159 ~/lab5 (0.174s)
sudo ./kernel_stack push 42

channel 21: open failed: connect failed: open failed

ubuntu@ip-172-31-27-159 ~/lab5 (0.174s)
sudo ./kernel_stack push 100

channel 21: open failed: connect failed: open failed

ubuntu@ip-172-31-27-159 ~/lab5 (0.173s)
sudo ./kernel_stack pop
100

channel 21: open failed: connect failed: open failed

ubuntu@ip-172-31-27-159 ~/lab5 (0.172s)
sudo ./kernel_stack pop
42

```

5.3 Simulating key removal

I used the sysfs control to remove the key without unloading the module:

```
$ sudo bash -c 'echo 0 > /sys/kernel/int_stack/control'
$ ls /dev/int_stack
ls: cannot access '/dev/int_stack': No such file or directory
$ ./kernel_stack pop
error: USB key not inserted
```

The device disappeared and the userspace utility correctly reported the absence of the key.

```
ubuntu@ip-172-31-27-159 ~/lab5 (0.179s)
sudo bash -c 'echo 0 > /sys/kernel/int_stack/control'

channel 21: open failed: connect failed: open failed
channel 23: open failed: connect failed: open failed
channel 25: open failed: connect failed: open failed

ubuntu@ip-172-31-27-159 ~/lab5 (0.178s)
ls /dev/int_stack
ls: cannot access '/dev/int_stack': No such file or directory

channel 21: open failed: connect failed: open failed
channel 23: open failed: connect failed: open failed
channel 25: open failed: connect failed: open failed

ubuntu@ip-172-31-27-159:~/lab5 (0.164s)
./kernel_stack pop
error: USB key not inserted
```

5.4 Simulating key reinsertion and verifying data persistence

I reinserted the key using sysfs and popped the remaining value:

```
$ sudo bash -c 'echo 1 > /sys/kernel/int_stack/control'
$ ls -l /dev/int_stack
crw----- 1 root root 239, 0 May 10 21:57 /dev/int_stack
$ sudo ./kernel_stack pop
42
```

The value 42 was pushed **before** the key was removed, proving that the stack data survived the entire removal/reinsertion cycle. The kernel log showed:

```
int_stack: key removed (simulated)
int_stack: key inserted (simulated)
```

```
ubuntu@ip-172-31-27-159 ~/lab5 (0.197s)
sudo ./kernel_stack push 42
sudo ./kernel_stack push 100
sudo ./kernel_stack pop
100

ubuntu@ip-172-31-27-159 ~/lab5 (0.18s)
sudo bash -c 'echo 0 > /sys/kernel/int_stack/control'

ubuntu@ip-172-31-27-159 ~/lab5 (0.181s)
ls /dev/int_stack
ls: cannot access '/dev/int_stack': No such file or directory

ubuntu@ip-172-31-27-159 ~/lab5 (0.168s)
./kernel_stack pop
error: USB key not inserted

ubuntu@ip-172-31-27-159 ~/lab5 (0.18s)
sudo bash -c 'echo 1 > /sys/kernel/int_stack/control'

ubuntu@ip-172-31-27-159 ~/lab5 (0.248s)
ls -l /dev/int_stack
crw----- 1 root root 239, 0 May 10 21:55 /dev/int_stack

ubuntu@ip-172-31-27-159 ~/lab5 (0.175s)
sudo ./kernel_stack pop
42
```

Requirements checklist

Requirement	Status
USB key gates the character device	Done
Real VID/PID used (058f:6387)	Done
Device appears on insertion, disappears on removal	Done
Stack data survives key removal	Done
<code>kernel_stack</code> reports error when key is missing	Done
Report with screenshots	Done

Key technical decisions

- **Real VID/PID** – I used 058f:6387, a documented Alcor Micro USB flash drive controller. The same values are used in the kernel's `usb_device_id` table and in the lecture examples.
- **Sysfs control for emulation** – because the `dummy_hcd + configs` gadget failed to enumerate, I implemented a simple sysfs knob that triggers the exact same code paths as a real USB event. This allowed me to conclusively demonstrate data survival.
- **Memory preservation** – `disconnect()` deliberately does not call `kfree(stack)`. The stack buffer and the current `top` remain untouched until the module is unloaded.

- **Default stack size** – `probe()` allocates a 128-element stack on first use, making the module immediately operational without a manual `set-size`.
-

Conclusion

Lab 5 was a success. I extended the Lab 4 kernel module with a USB key gate, using real USB identifiers and a fully functional probe/disconnect lifecycle. The character device appears and disappears in response to key events, and the internal stack data persists when the key is removed and reinserted. The userspace utility was updated to correctly report the key status. All testing was performed on Ubuntu 22.04 under AWS, with a virtual USB controller and a custom sysfs trigger to overcome emulator limitations.